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Motor recovery from constraint induced movement therapy is not constrained by extent of tissue damage following stroke

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Abstract.
Purpose: The purpose of this study was to delineate the relationship between several types of T1-weighted MRI pathology and motor rehabilitation potential following Constraint Induced Movement therapy (CI therapy) in chronic stroke.
Methods: Stepwise regression was employed ($n = 80$) to identify predictors of motor recovery (prior to therapy) and of response to Constraint-Induced Movement therapy [measured via the Wolf Motor Function Test (WMFT) and Motor Activity Log (MAL)] from among the following: age, side of motor deficit, chronicity, gender, lesion volume, peri-infarct damage volume, white matter hypointensity volume, ventricular asymmetry, and lesion location.
Results: Although extent of total stroke damage weakly correlated with poorer performance on the WMFT prior to therapy, this relationship was mediated by the location of the damage. No metric of tissue damage examined here was associated with real-world arm use at baseline (MAL at pre-treatment) or with CI therapy-induced improvement in either best motor performance upon request (WMFT) or spontaneous arm use for daily activities (MAL).
Conclusions: In sum, the *extent* of brain tissue damage of any type examined here poorly predicted motor function and response to rehabilitation in chronic stroke.

Keywords: Stroke, rehabilitation, CI therapy, lesion volume, infarct volume, motor

1. Introduction

Motor rehabilitation potential following stroke is anticipated to relate to the size and functional anatomic location of injury. Although there is fairly consistent evidence that lesion *location* influences subsequent recovery of motor function (Gauthier et al., 2009; Lindenberg et al., 2010; Lo et al., 2010; Zhu et al., 2010), the literature portrays an inconsistent and overall weak relationship between lesion volume and motor ability in both acute and chronic stroke (Lauretani et al., 2010; Lindberg et al., 2007; Mark et al., 2008b; Mori et al., 1994; Page et al., 2013; Riley et al., 2011; Sterr et al., 2010).

The largest lesion study ($n = 513$), which followed individuals from the acute phase through one-year follow-up, showed that total anterior circulation

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infarcts resulted in greater initial impairment (lower Barthel index) and longer periods of rehabilitation than partial or lacunar anterior circulation infarct subtypes, but extent of recovery was indistinguishable between subtypes by the chronic phase (Lauretani et al., 2010). Another large sample study ($n = 300$) found that infarct volume only weakly predicted chronic motor outcomes (as measured by Brunstromm staging, Mori et al., 1994). These studies were limited, however, by imprecise measurement of motor ability (e.g., Brunnstrom staging, Barthel index) (Lauretani et al., 2010; Mori et al., 1994), and categorical classification of lesion size (Lauretani et al., 2010).

A number of smaller-sample studies employed more sensitive measures of motor function and lesion volume. The great majority of these did not show a meaningful relationship between lesion volume and spontaneous motor recovery (Lindberg et al., 2007; Mark et al., 2008b; Page et al., 2013; Riley et al., 2011; Sterr et al., 2010) or response to rehabilitation in chronic stroke (Mark et al., 2008b; Riley et al., 2011; Sterr et al., 2010), albeit with some exceptions (Crafton et al., 2003; Cramer et al., 2007). Consensus across studies may be limited by the very small sample sizes employed in some investigations (Crafton et al., 2003; Cramer et al., 2007; Lindberg et al., 2007; Riley et al., 2011; Sterr et al., 2010), different measures of motor function, and inconsistent inclusion/exclusion criteria between studies. A shared limitation of this prior work, however, is the exclusion of other MRI pathology from investigation.

Given the diverse pathology following stroke, concurrent examination of several types of MRI pathology, in conjunction with lesion location, is needed to clarify the relationship between tissue pathology and motor function. For example, the accuracy of measurements of lesion volume in the chronic phase may be compromised by marked atrophy of the damaged parenchyma (Bigler and Maxwell, 2011) and accompanying ex vacuo ventricular enlargement. It is thus important to include extent of hydrocephalus ex vacuo in the analysis. Another type of (often overlooked) tissue damage is reduced tissue integrity surrounding the infarct. Preserved tissue adjacent to the infarct often has diminished integrity and exhibits abnormal metabolism (Beaulieu et al., 1999). Although this tissue remains somewhat structurally intact, it is unclear to what extent it is functioning effectively, and thus to what extent this damage contributes to lasting motor deficit. A third type of commonly observed tissue damage not previously accounted for in studies relating MRI pathology to motor function is reduced white matter integrity secondary to small vessel disease. Small vessel disease is a common risk factor for stroke and often produces tissue damage that can be captured via MRI (e.g., hypointensities in white matter on T1-weighted images or FLAIR hyperintensities, most commonly observed in periventricular regions). It is currently unknown whether this damage increases total lesion burden in individuals who have experienced stroke. Finally, predictive relationships between brain tissue damage and motor function found in some studies (Chen et al., 2000; Crafton et al., 2003; Cramer et al., 2007) may be better explained by lesion location effects, but few studies have examined these factors together.

Examined here is the relationship between several different types of pathological MRI findings on motor recovery following usual care and response to a specific, carefully controlled form of rehabilitation (CI therapy). The latter has particular basic significance because the effectiveness of CI therapy has been linked to neuroplasticity, a process that could theoretically be constrained by inflammatory response in surviving damaged tissue (Hughes et al., 2010; Jin et al., 2014; Price et al., 2006). Prior work shows that CI therapy has the potential to alter brain structure (Gauthier et al., 2008; Sterling et al., 2013), cortical excitability (Liepert et al., 2000), neurogenesis and gene expression (in animal models) (Maier et al., 2008; Schneider et al., 2014; Zhao et al., 2009). Moreover, grey matter density in *nondamaged* regions has been shown to predict CI therapy response (Gauthier et al., 2012). These findings raise the question of whether a greater extent of damage to non-infarcted regions could interfere with the brain's ability to reorganize, and hence the effectiveness of CI therapy.

Based on previous work that showed large clinical gains from CI therapy irrespective of damage to the corticospinal tract (Rickards et al., 2014), we hypothesized that rehabilitation response would be impervious to pathology in remaining brain tissue (e.g., hydrocephalus ex vacuo and peri-infarct tissue damage). Alternatively, extensive damage to non-infarcted tissue could interfere with CI therapy response by reducing the potential for neuroplasticity, a likely mechanism for the effectiveness of CI therapy (Gauthier et al., 2008; Liepert et al., 2000). Utilizing retrospective imaging data acquired during several previous clinical trials (Taub et al., 2005, 2006, 2013), this study is the first

to concurrently model the effects of several different types of pathological MRI findings on motor ability in the chronic phase of stroke and response to CI therapy.

2. Methods

2.1. Participants

Participants consist of 80 individuals (40 females, 40 males) with chronic stroke and mild-moderate upper extremity motor deficit who were enrolled from 1997 to 2009 in various comparative effectiveness trials conducted at the University of Alabama at Birmingham and for whom good quality volumetric MRI data was obtained. Participants had been randomized to receive either Constraint-Induced Movement therapy (CI therapy) or alternative (control) interventions (Taub et al., 2005, 2006, 2013). Individuals were recruited without respect to lesion etiologies (e.g., cortical vs. subcortical, ischemic vs. hemorrhagic infarcts, lacunes) to maximize the external validity of the findings. All participants provided signed informed consent to participate. Inclusion criteria consisted of: post-stroke interval of more than 10 months; ability to initiate wrist extension, thumb abduction, and extension of at least 2 additional fingers; substantial nonuse of the more-affected arm (i.e., score $\leq$2.5 out of 5 on the Motor Activity Log), general medical stability, ability to follow basic verbal commands, absence of current pharmacological treatment of their motor disability (e.g., no botulinum toxin injections within the past 3 months), and no history of having received Constraint-Induced Movement therapy.

Participants received volumetric T1-weighted MRI of their brain prior to receiving any study-related intervention. The most recent participants ($n = 49$) were scanned on a Phillips (Intera) 3 Tesla machine, less recent participants on either a Philips (ACS) 1.5 Tesla machine ($n = 14$) or a GE (Signa) 1.5 Tesla machine ($n = 17$). Image acquisition parameters for the GE 1.5 Tesla MRI included axial 3D spoiled gradient recalled (SPGR) acquisition with 1.5 mm slice thickness, in-plane resolution of 0.90 mm $\times$ 0.90 mm, repetition time (TR) = 30, echo time (TE) = 6, field of view (FOV) = 250, flip angle (α) = 45. Acquisition parameters for the Philips 1.5 Tesla were identical except TR = 20 and $\alpha = 8$. Image acquisition parameters for the Phillips 3 Tesla MRI included axial 3D Turbo-Field Echo (TFE) acquisition with 1 mm slice thickness, in-plane resolution of 0.94 mm $\times$ 0.94 mm, TR = 9.8, TE = 4.6, FOV = 250, $\alpha = 8$. T2 FLAIR images were also obtained for a subset of participants ($n = 38$). For the 3 Tesla MRI, the FLAIR sequence was acquired axially in 4 mm slices with 1 mm gaps, TR = 11,000, TE = 120, inversion time (TI) = 2,800, $\alpha = 90$. FLAIR parameters for the GE 1.5 T were axial 4 mm slices with no gap, TR = 9,002, TE = 142.5, TI = 2,200, $\alpha = 90$. Participants ranged in age from 38 to 87 years (mean, 62.9 $\pm$ 12.3). Stroke onset was 4.3 $\pm$ 4.0 years previously. Thirty-nine exhibited right hemiparesis; 71 were right-handed prior to stroke. See Table 1 for a breakdown of patient characteristics.

2.2. Assessment of pre-treatment motor functioning

Motor functioning was assessed before and after treatment with the Motor Activity Log Arm Use scale (MAL) and the Wolf Motor Function Test (WMFT), both of which have established reliability and validity. The MAL quantifies how well and how often activities of daily living are carried out *spontaneously* using

Table 1
Patient characteristics prior to CI therapy ($n = 80$)

	Mean $\pm$ SD	Min	Max
Age	62.9 $\pm$ 12.3	38	87
Chronicity (years)	4.3 $\pm$ 4.0	0.8	20
Percentage with right hemiparesis	48.8		
Percentage of purely subcortical infarcts	51.3		
Infarction volume (% of ICV)	0.8 $\pm$ 1.7*	0.002	9.1
Stroke damaged volume (% of ICV)	1.8 $\pm$ 2.6*	0.002	11.9
White matter hypointensity volume (% of ICV)	0.2 $\pm$ 0.4*	0.0	2.0
Ventricular asymmetry ratio (1 = ventricles of equal volume)	1.4 $\pm$ 0.6*	0.7	3.8
Motor Activity Log Quality of Movement scale (0–5 points)	1.0 $\pm$ 0.7	0.0	2.6
Wolf Motor Function Test Log^2 Performance Time	1.3 $\pm$ 1.0	0.1	3.8

*positively skewed distribution, ICV = intracranial volume.

the more-affected arm *outside the laboratory* (Taub et al., 1993; Uswatte et al., 2006; van der Lee et al., 2004). The WMFT measures how rapidly participants perform standardized tests that they are *required* to carry out with the more-affected arm *inside* the laboratory (Morris et al., 2001; Wolf et al., 2001). The two measures thus assess distinct domains of motor ability (Taub et al., 2006; Uswatte and Taub, 1999). Performance time on the WMFT was recorded as a $\log^2$ transformation of the mean time in seconds to more accurately portray patient progress (Mark et al., 2008b).

2.3. Interventions

Forty-three of 80 participants received CI therapy, a motor rehabilitation therapy with strong evidence of efficacy for treating hemiparesis in chronic stroke (Morris et al., 2006; Taub et al., 1993, 2006; Wolf et al., 2006). The other participants were randomly assigned to other interventions. Only those participants assigned to CI therapy ($n = 43$), the most effective intervention, were retained for the analysis examining predictors of therapy outcome. The demographics of those receiving CI therapy were similar to those of the entire sample. Participants receiving CI therapy had a mean age of 62.1 ± 12.2 years and mean stroke chronicity of 4.3 ± 3.9 years; 23 were women, 22 exhibited right hemiparesis, and 41 were right-handed prior to stroke.

CI therapy is characterized by at least 30 hours of high-repetition in-clinic training of the more impaired arm on functional tasks, restraint of the less-impaired arm for a target 90% of waking hours, and a number of behavioral techniques termed the "transfer package" lasting an additional 0.5 hours in the laboratory (Taub et al., 1999). The transfer package, designed to facilitate transfer of therapeutic gains to real world activities, includes daily monitoring of life situation use of the more affected arm in several ways, problem-solving with a therapist to overcome perceived barriers to using the extremity, and a behavioral contract, among other techniques (Morris et al., 2006; Taub et al., 2013). Although some characteristics of the CI therapy protocol differed across studies (e.g., sling versus mitt restraint), comparable treatment outcomes were observed across studies (Taub et al., 1999).

2.4. MRI Analysis

The following types of tissue damage were quantified on T1-weighted MRI images: *infarcts* [cerebrospinal fluid (CSF) filled space left behind by stroke], *peri-infarct damage*, and (non-CSF) *white matter hypointensities*. Brain voxels with T1-weighted hypointensity were classified either as peri-infarct tissue damage when no normal-appearing tissue was present between the CSF-filled infarct and the voxels in question, or as non-specific white matter hypointensity when it appeared remote from an infarct. *Infarcts* (cerebrospinal fluid filled space left behind by stroke), *peri-infarct damage*, and *white matter hypointensities* were separately traced by four blinded raters under the supervision of a neurologist (V.W. Mark) using MRIcro/MRIcron software (see Fig. 1 for delineation of these regions). Tracings were then refined using semi-automated intensity thresholding (of only the traced region). The thresholding procedure involved identifying an intensity cutoff that best distinguished damaged from visually-intact tissue and applying this threshold to the entire manual trace. Although this thresholding step did not greatly alter the resultant volumes, it expedited tracing [by allowing for less precise (slightly over-inclusive) delineation of the lesion border] and reduced the potential for human error. Each rater performed tracings on a minimum of 20 scans. Interrater and intrarater reliability of the thresholded tracings was excellent (>.99), as assessed via intraclass correlation on an overlapping subset of 10 images for each rater or rater-pair. Individuals performing the tracing were unaware of participants' motor status. To control for variability in brain size, thresholded tracings of damaged tissue were expressed as a percentage of

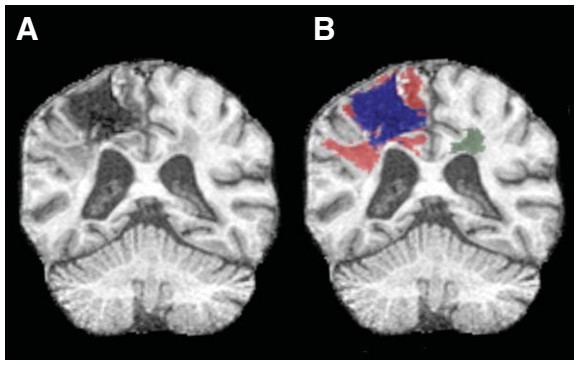


Fig. 1. Several different types of damage were quantified within each individual's scan. A: Coronal reconstruction from an original axial acquisition. B: The infarct is depicted in blue, peri-infarct tissue damage in red, and white matter hypointensities in green. Note also the size discrepancy between the right ventricle (adjacent to the lesion) and the contralateral ventricle.

intracranial volume. Intracranial volume was calculated automatically by 1) skull-stripping the images (during which lesions were masked to avoid misclassifying cortical lesions), 2) summing the number of voxels that remained above the level of the medulla, and 3) multiplying by the voxel size. A topographic histogram (i.e. damaged tissue overlap map) depicting the voxel-wise prevalence of infarct, peri-infarct damage, and white matter hypointensities combined (normalized to a custom older adult template (Lemaitre et al., 2005)) is shown in Fig. 2. FLAIR imaging is often utilized as the clinical standard for quantifying white matter pathology due to its ability to more conspicuously capture infarctions and white matter disease (Alexander et al., 1996). Because the equivalence of different MRI metrics for quantifying white matter pathology (e.g., FLAIR hyperintensities versus T1-weighted hypointensities) has not yet been established, *white matter hyperintensities* and *periventricular hyperintensities* were also quantified using Fazekas ratings (Fazekas et al., 1987, see Table 2) for a subset of 38 participants for which non-volumetric FLAIR images were available. See Fig. 3 for a direct comparison of MRI pathology visualized via T1-weighted and FLAIR imaging. Fazekas rating scale has been shown to be significantly correlated with white matter hyperintensity volumetric calculation (Gouw et al., 2006). Lesion distortion resulting from focal neuronal degeneration (and accompanying ventriculomegaly) was indirectly quantified by calculating the extent of *ventricular asymmetry* - the ratio of the volume of the lateral ventricle contralateral to the motor deficit relative to the opposite ventricle – on the participants' original (not normalized) MRI scans.

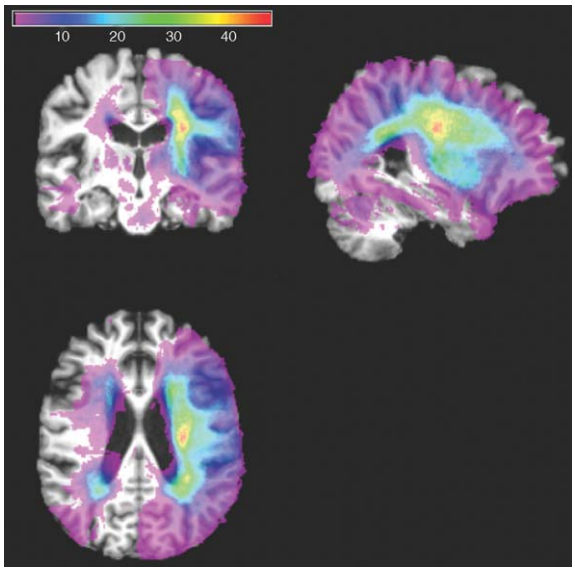


Fig. 2. Topographical brain map depicting the number of participants with apparent tissue damage at each voxel. Brains of participants with left hemiparesis were flipped right-to-left to equate for deficit side. Warmer colors (red and yellow) indicate brain regions in which more participants exhibited tissue damage. The maximal overlap occurs in the corona radiata, with tissue damage prevalent in 59% (47/80) of subjects.

Table 2
Description of Fazekas ratings

	Rating	Description
Periventricular hyperintensities	0	absence
	1	"caps" or pencil-thin lining
	2	smooth "halo"
	3	irregular hyperintensities extending into the deep white matter
Deep white matter hyperintensities	0	absence
	1	punctate foci
	2	beginning confluence of foci
	3	large confluent areas

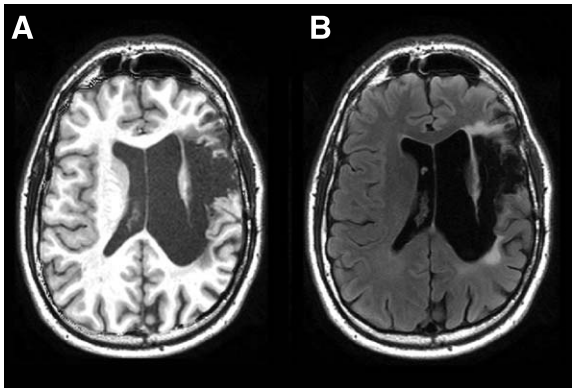


Fig. 3. Direct comparison of MRI pathology visualized via T1-weighted MRI versus FLAIR. A: T1-weighted MRI for one participant coregistered to FLAIR. B: FLAIR MRI for the same participant.

2.5. Statistical Analysis

Bidirectional stepwise regression (entrance and exit tolerances of .05 and .10, respectively) was performed using Matlab's statistical toolbox to model the effects of damaged tissue extent on motor outcomes at pretreatment ($n = 80$). Two separate models were generated, with WMFT and MAL as the outcome

measures, respectively, age, side of motor deficit, chronicity, and gender as potential nuisance covariates, and lesion volume, total stroke damage volume, T1-weighted white matter hypointensity volume, and ventricular asymmetry as potential predictor variables. Where significant effects were found, lesion location (whether or not the stroke-damaged region overlapped with the intersection of the corona radiata and corpus callosum - a region found to be associated with greater impairment on the WMFT in this sample of participants (Gauthier et al., 2009) and linked to greater motor impairment in other studies (Lindenberg et al., 2012; Lo et al., 2010) was tested as a potential mediating variable through use of the Sobel procedure.

To ensure consistent results when white matter damage is quantified via FLAIR hyperintensities versus T1-weighted hypointensities, a similar bidirectional stepwise regression was run incorporating Fazekas periventricular hyperintensity ratings and Fazekas deep white matter hyperintensity ratings in lieu of T1-weighted white matter hypointensity volume for the subset of 38 participants for which Fazekas scores could be obtained. Fazekas ratings for both periventricular hyperintensity and deep white matter hyperintensity were dichotomized in the model (ratings of 0 and 1 were grouped together and ratings of 2 and 3 were grouped together), given the low prevalence of 0 and 2 Fazekas ratings in our dataset.

To examine the effects of damaged tissue on response to CI therapy, identical stepwise regression procedures were used, with the exception that post-treatment WMFT or MAL score served as the outcome measure, respectively, and pre-treatment WMFT or MAL score served as an additional covariate. The initial model was constrained such that pre-treatment WMFT or MAL score entered into the model first. See Table 3 for a depiction of the statistical models used.

The positively skewed volumetric measurements in this dataset violated some parametric assumptions (non-normal distribution of residuals for white matter hypointensity volume and chronicity). Bootstrap analysis revealed that the resultant models were not significantly affected by these violations.

3. Results

3.1. Clinical response to CI therapy

CI therapy recipients showed large significant improvements on the MAL ($t_{(42)} = 15.89$; $P < 0.0001$; d' = 2.41) and the WMFT ($t_{(42)} = 4.04$; $P = 0.0002$; d' = 0.62).

3.2. MRI predictors of motor performance

Consistent with prior work (Taub et al., 1999, 2006; Wolf et al., 2006), age, side of greatest motor impairment, chronicity, and gender were not significant predictors in any of the models. The model relating characteristics of tissue damage to WMFT performance at baseline yielded two significant predictors: total stroke damage volume ($p < 0.004$) and infarct volume ($p = 0.023$, see Table 3). Participants with larger total stroke damage performed significantly slower on the WMFT. It should be noted, however, that less than 10% of the variance in WMFT scores (Adjusted $R^2 = 0.098$) was accounted for by volume of stroke-damaged tissue. Furthermore, when location of damage was added to the model, it was found to fully mediate effects that were initially attributed to volume of damage [i.e. greater volume of tissue damage was associated with lesions at the intersection of the corona radiata and corpus callosum ($p < 0.001$), lesions in this location predicted poorer motor function ($p = 0.014$), and lesion location replaced volume of stroke-damaged tissue as the sole predictor of WMFT performance when entered into the full model ($p = 0.014$, Adjusted $R^2 = 0.063$)]. Similar results were obtained when Fazekas scores were utilized in lieu of peri-infarct and T1-weighted white matter hypointensity volume on a subset of 38 participants. More severe deep white matter hypointensity Fazekas ratings (largely influenced by the extent of damaged tissue surrounding the infarct) predicted poorer WMFT scores ($p = 0.005$), but this relationship disappeared entirely after accounting for lesion location. The consistency in results regardless of how tissue damage was quantified (i.e., via Fazekas score or volume traced on T1-weighted scan) is not surprising given excellent correspondence between the two metrics [$t = 5.8$, $p < 0.0001$, effect size = 2.6 for comparison of T1-weighted white matter hypointensity volume between those who received lower (<2) and higher Fazekas scores (≥ 2)]. These results indicate that regardless of how damaged tissue is measured (i.e., using T1-weighted or FLAIR images), volume of stroke-damaged tissue is not a useful predictor of motor dysfunction after accounting for more salient predictors (lesion location). Extent of tissue damage (irrespective of how it was quantified) was not associated with real-world arm use at baseline

Table 3
Statistical step-wise regression models

Outcome variables	Predictor Variables	β, t	Sobel β, t
Motor Activity Log (MAL)	Age	N.S.	–
Quality of Movement scale	Gender	N.S.	–
at baseline	Chronicity	N.S.	–
	Side of motor deficit	N.S.	–
	Infarction volume	N.S.	–
	Total damaged volume	N.S.	–
	White matter hypointensity volume	N.S.	–
	Ventricular assymetry ratio	N.S.	–
Wolf Motor Function Test (WMFT) Log^2	Age	N.S.	N.S.
Performance Time at baseline	Gender	N.S.	N.S.
$F = 5.29$	Chronicity	N.S.	N.S.
$P = 0.007$	Side of motor deficit	N.S.	N.S.
Adjusted $R^2 = 0.098$	Infarction volume	−40.9, −2.31	N.S.
Added lesion location:	Total damaged volume	33.6, 2.99	N.S.
$F = 6.34$	White matter hypointensity volume	N.S.	N.S.
$P = 0.014$	Ventricular assymetry ratio	N.S.	N.S.
Adjusted $R^2 = 0.063$	*Lesion location	–	0.58, 2.52
Motor Activity Log Quality of	Baseline MAL	0.82, 4.48	–
Movement scale at post-treatment	Age	N.S.	–
	Gender	N.S.	–
	Chronicity	N.S.	–
	Side of motor deficit	N.S.	–
	Infarction volume	N.S.	–
	Total damaged volume	N.S.	–
	White matter hypointensity volume	N.S.	–
	Ventricular assymetry ratio	N.S.	–
Wolf Motor Function Test Log^2	Baseline WMFT	0.82, 13.05	–
Performance Time at post-treatment	Age	N.S.	–
	Gender	N.S.	–
	Chronicity	N.S.	–
	Side of motor deficit	N.S.	–
	Infarction volume	N.S.	–
	Total damaged volume	N.S.	–
	White matter hypointensity volume	N.S.	–
	Ventricular assymetry ratio	N.S.	–

*Lesion location was added to the model during the Sobel procedure to test for mediation.

(MAL at pre-treatment) or with CI therapy-induced improvement in either motor performance (WMFT) or real-world arm use (MAL). In sum, the extent of brain tissue damage of any type does not appear to relate to motor function or response to rehabilitation in chronic stroke.

4. Discussion

By concurrently evaluating several types of chronic stroke MRI pathology, in conjunction with lesion location, this study represents a more thorough analysis of the relationship between tissue damage and recovery of motor function. Results show that extent of tissue damage of any type does not relate to either baseline motor function or CI therapy response, when controlling for the location of this damage. These results are consistent with most previous studies that examined necrotic volume in isolation (Lindberg et al., 2007; Mark et al., 2008b; Riley et al., 2011; Sterr et al., 2010; Zhu et al., 2010), and show that little predictive value is captured by adding other common types of tissue damage to the model.

Furthermore, these results suggest that the weak predictive relationships between brain tissue damage and motor function found in some studies (Chen et al., 2000; Crafton et al., 2003; Cramer et al., 2007) may be confounded by lesion location effects. This is consistent with other findings in the literature. For example, lesion volume within each region of interest is a much stronger predictor of motor performance

than lesion volume in isolation (Chen et al., 2000). In other accounts, lesion volume appears to lose its predictive validity when extent of lesion overlap with the corticospinal tract or the motor map or is accounted for (Crafton et al., 2003; Zhu et al., 2010).

Although lesion location is a more robust predictor of motor ability in the chronic phase of stroke than volume of tissue damage, it should be noted that it still only explains a small portion of the variance in motor status (less than 8%). This is consistent with the hypothesis that the brain compensates for the effects of the initial damage by harnessing alternative motor pathways, thereby lessening the impact of the initial infarct. This hypothesis garners support from prior research that showed that significant recovery can occur by the chronic phase even with complete disruption of the corticospinal tract, despite greater deficits initially (Jang et al., 2008; Swayne et al., 2008). Moreover, the predictive validity of corticospinal disruption on motor status diminishes over time, while the inverse is true of alternate motor fibers (Lindenberg et al., 2010; Swayne et al., 2008). Also in support of this hypothesis is the observation that infarcts impacting multiple motor pathways have a more lasting detrimental effect (Fries et al., 1993; Lindenberg et al., 2010).

The present investigation builds substantially upon prior lesion studies in that it additionally investigated the effect of different types of brain damage on individuals' ability to benefit from CI therapy. CI therapy is an intensive intervention for hemiparesis secondary to stroke, TBI, cerebral palsy, and multiple sclerosis (Choudhary et al., 2012; Mark et al., 2008a; Shaw et al., 2005; Sterling et al., 2013; Taub, 2004, 1993) that is known to promote neuroplasticity (Gauthier et al., 2008; Liepert et al., 2000; Sterling et al., 2013). As such, it provides an excellent model for evaluating clinical response/ rehabilitation potential (because the majority of patients show large clinical gains in response to the therapy) (Wolf et al., 2006). However, despite its wide presence in the literature, there exist no published accounts relating the different types of brain damage studied in this paper to CI therapy outcome. The finding that CI therapy outcome was independent of (all types of) MRI pathology measurable via routine clinical scans suggests that rehabilitation-induced neuroplasticity is unaffected by both lesion extent and extensive damage to non-infarcted tissue.

It has been hypothesized that similar neural recovery mechanisms operate during a targeted intervention as during recovery in the absence of intervention (Cramer, 2008). The present findings, taken together with other accounts in the literature, support an alternate view that CI therapy targets different or additional neurorecovery processes. Reported here is the finding that location of stroke damage predicted motor ability prior to receiving CI therapy (recovery in the chronic phase following usual care), but not response to CI therapy, evidence that the neural constraints on recovery following usual care can be overcome by treatment with CI therapy. A similar dissociation was observed by Rickards and colleagues when examining the effect of disruption of the corticospinal tract on both initial motor status and response to CI therapy (Rickards et al., 2014). The inverse has also been shown whereby grey matter density in brain regions that are not predominantly associated with motor function predicted CI therapy response, but not pre-treatment motor ability. The combination of these findings reveals a double-dissociation that raises the possibility that CI therapy harnesses different neural substrates than recovery following usual care. It appears that the recovery mechanisms harnessed by CI therapy are not hindered by the initial damage that produced the motor deficit, but are dependent, in part, on the density of undamaged cortical tissue (Gauthier et al., 2012).

This study has a few limitations that are worthy of mention. A retrospective dataset of patients who were recruited for CI therapy treatment studies was utilized. As such, conventional MRI acquisition sequences were employed on the select group of chronic stroke patients that met enrollment criteria for these CI therapy studies, namely patients with mild to moderate hemiparesis. MRIs from this dataset were obtained using only commonly available scanning strengths and acquisition sequences, potentially limiting the sensitivity of our analyses. However, given that extent of tissue damage was not even marginally predictive of motor function after controlling for lesion location (ps >0.32), it is unlikely that enhanced sensitivity (e.g., from high-resolution FLAIR imaging) would impact our overall findings. A final limitation of these findings was that participants had infarcts of various etiologies that could not be elucidated by the research team (given that individuals with chronic stroke were recruited for this study), and thus could not be controlled for in data analysis. To our knowledge, no data exists regarding whether stroke etiology results in chronic tissue differences between hemorrhagic versus ischemic lesions. Additional research is needed to determine whether lesion etiology may impact these findings.

In sum, extent of tissue damage is a relatively unsophisticated predictor of motor dysfunction that carries little value. Lesion location (Gauthier et al., 2009; Lindenberg et al., 2010; Lo et al., 2010; Zhu et al., 2010) and integrity of remaining tissue (Gauthier et al., 2012) have shown more promise for predicting lasting motor deficit; however, these measures are still unable to quantify recovery trajectories of individual patients or the extent to which motor networks as a whole are affected. The motor network is a complicated system of disparate brain regions that work in concert to execute motor movements. Research studying network-based models of motor deficit is thus more likely to yield a greater depth of knowledge than can be gained by studying facets of damage in isolation. For example, neuroplasticity may allow for recovery of function after stroke by strengthening alternative pathways (Zaaimi et al., 2012). Recovery of function may therefore depend in large part on the viability of remaining pathways and the extent to which they are able to reproduce the lost function (Lindenberg et al., 2012). More research along this line is needed to better understand the effects of stroke on interconnected components of our complex motor circuitry and, ultimately, motor function.

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